

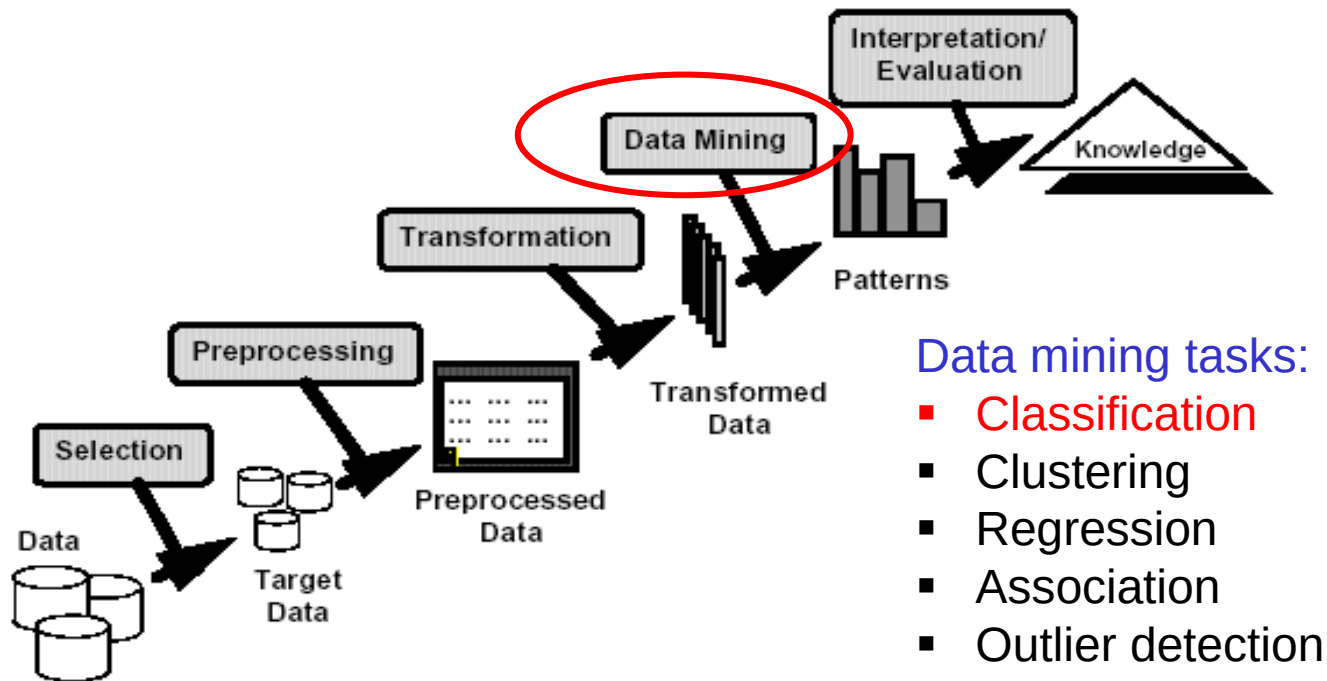
Lecture 4:

Data classification (II)

= Classification trees and classification rules =

Motivation

Reminder: knowledge discovery workflow



Reminder

- Classification models:
 - Rules based models
 - Instance based models
 - Probabilistic models
 - Function based models
- Usage of the data:
 - Training dataset – used to construct the model
 - Validation – used to set the hyper-parameters of the model
 - Testing – used to evaluate the performance of the model
- Performance evaluation:
 - Confusion matrix
 - Performance measure: accuracy, sensitivity/specificity, precision / recall, F-score, AUC (area under receiver-operator-characteristic curve)

Outline

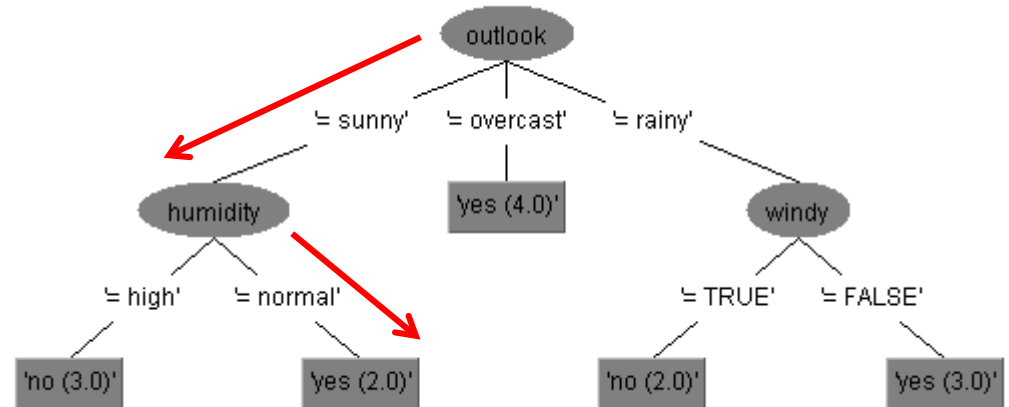
- Classification models
 - Decision trees
 - Splitting criteria
 - Recursive splitting (ID3 algorithm)
 - Pruning
 - Decision rules
 - Characteristics of rulesets
 - Rules support and confidence
 - Covering algorithms (RIPPER)

Decision Trees

Weather/play dataset

Decision tree (constructed using Weka)

Relation: weather.symbolic					
No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no



How can be used?

Which class corresponds to a new instance
(outlook=sunny, temperature=mild, humidity=normal, windy=False)?

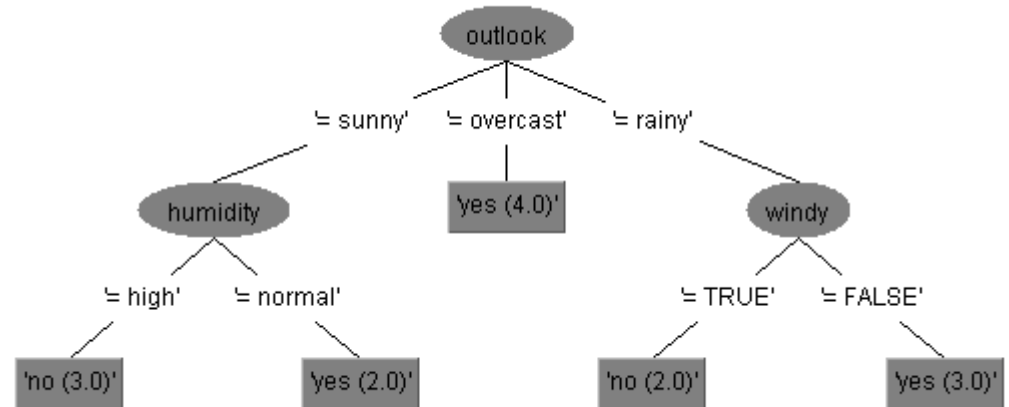
Class: **Yes**

Decision Trees

Weather/play dataset

Decision tree (constructed using Weka)

Relation: weather.symbolic					
No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no



How can be translated in a set of classification rules? Each branch leads to a rule

- Rule 1:** IF outlook=sunny and humidity=high THEN play=no
- Rule 2:** IF outlook=sunny and humidity=normal THEN play=yes
- Rule 3:** IF outlook=overcast THEN play=yes
- Rule 4:** IF outlook=rainy and windy=True THEN play=no
- Rule 5:** IF outlook=rainy and windy=False THEN play=yes

Decision Trees

How can a decision tree be constructed (inferred) from a dataset?

- **Choose an attribute** and put it in the root of the tree
- For **each possible value** of the attribute (present in the dataset) **construct a branch** (split the node)
- **Split the dataset** in subsets corresponding to each branch
 - If a subset contains data from only one class then it will correspond to a leaf node (no more splitting on that branch) – **pure node**
 - If in a subset there are data belonging to different classes then the splitting process is continued until
 - is arrived to a pure node
 - all attributes have been analyzed on that branch
 - the subset of data corresponding to that branch is empty

Weather/play dataset

Relation: weather.symbolic					
No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

Questions: in which order should be analyzed the attributes? which test condition should be assigned to a node?

Decision Trees

In which order should be analyzed the attributes?

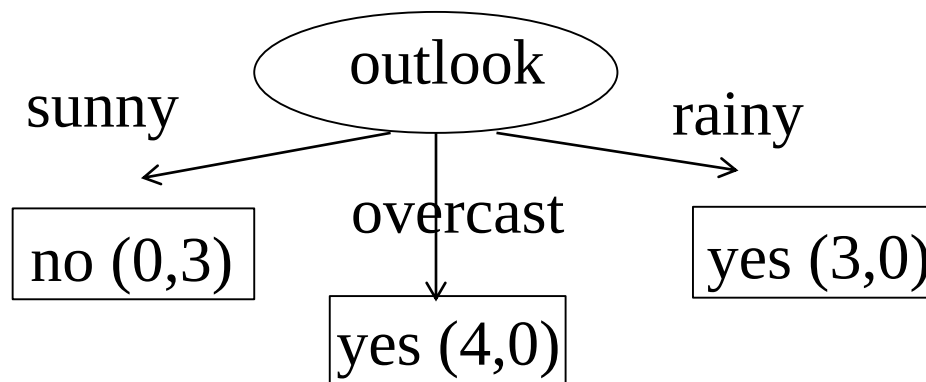
Weather/play dataset (selected instances – just to illustrate the idea)

Main idea:

- Select the attribute which leads to a simple tree, i.e. an attribute with a high purity level (ideally, for each possible value of the attribute the corresponding data instances belong to the same class)

No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	overcast	cool	normal	TRUE	yes
7	sunny	mild	high	FALSE	no
8	rainy	mild	normal	FALSE	yes
9	overcast	mild	high	TRUE	yes
10	overcast	hot	normal	FALSE	yes

Example:



Remark:

- All leaves are “pure” (contain data belonging to the same class)
- Such a flat tree leads to classification rules involving only one attribute in the left-hand side (similar to OneR)
- This situation happens rarely for real-world data

Decision Trees

The main decisions to be taken during the decision tree induction

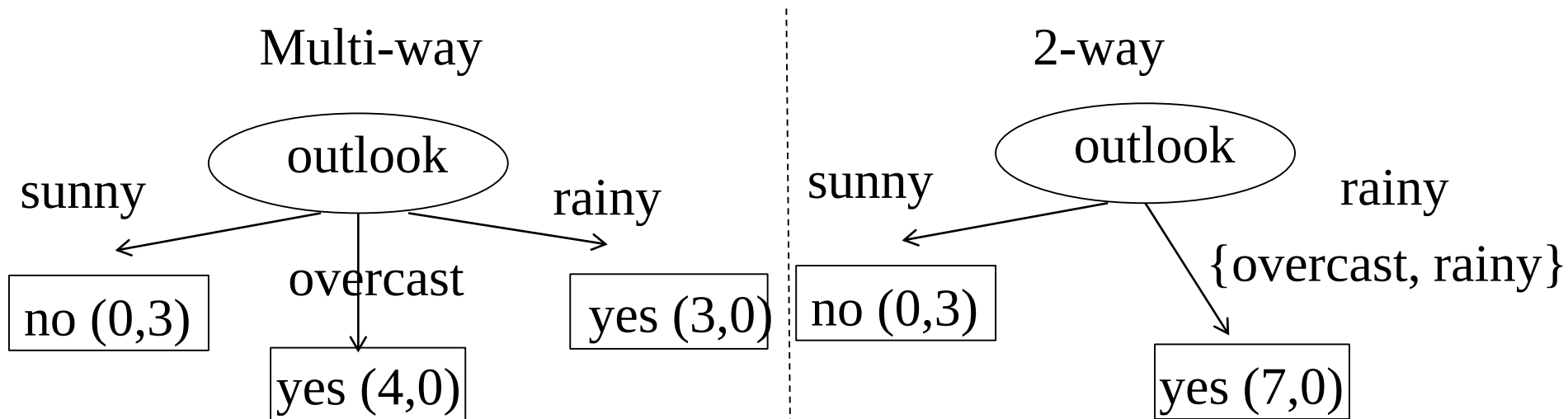
- Which attribute should be selected for splitting?
 - The most discriminative one – that which ensures a partition of the current dataset in subsets with a high degree of purity
 - There are several criteria which can be used:
 - Entropy (variants: information gain, gain ratio)
 - Gini index
 - Misclassification score
- Which are the test conditions to be assigned to the branches corresponding to a node?
 - It depends on the attributes type
 - Nominal, ordinal: $\text{attribute} = \text{value}$
 - Continuous: $\text{attribute} < \text{value}$ or $\text{value1} < \text{attribute} < \text{value2}$
 - It depends on the desired degree of the splitting node:
 - 2-way split (the current subset is divided in 2 subsets) → binary tree
 - Multi-way split (the current subset is divided in several subsets) → multiway tree

Decision Trees

- Which are the test conditions to be assigned to the branches corresponding to a node?

Nominal and ordinal attributes:

- Multi-way: as many branches as possible values
- 2-way: two branches



Decision Trees

Which are the test conditions to be assigned to the branches corresponding to a node?

Numerical attributes:

- The numerical attributes are previously discretized and then is applied the approach which is specific to attributes with discrete values (using the multi-way or 2-way approach)

Which attribute should be selected for splitting?

- That which leads to the maximal reduction in the information needed to take the right decision
 - $\text{Information gain} = \text{Entropy}(\text{distribution of data before splitting}) - \text{AveragedEntropy}(\text{distribution of data after splitting})$

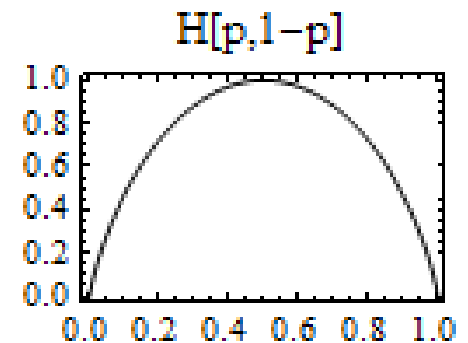
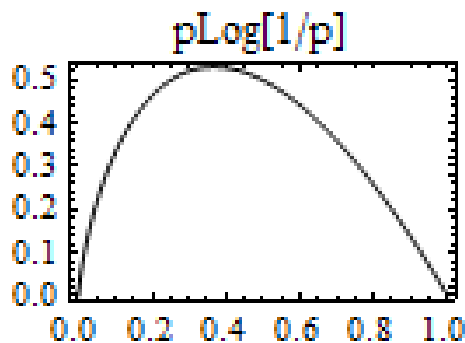
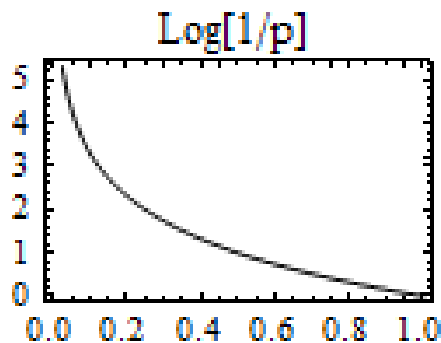
Reminder: entropy

Let $D=(p_1, p_2, \dots, p_k)$ be a distribution probability. The entropy associated to this distribution probability is defined as

$$H(D) = H(p_1, p_2, \dots, p_k) = -\sum_{i=1}^k p_i \log p_i$$

And can be interpreted as a **measure of the amount of uncertainty** (or surprise) when we sample data according to the distribution probability

Particular case: $k=2 \Rightarrow p_1=p, p_2=1-p$



Remark:

Log[1/p] interpretation: the surprise of seeing an event characterized by a small probability (unexpected event) is higher than in the case of an event of high probability (which is expected)

Reminder: entropy

In the context of a classification problem:

- $D=\{C_1, C_2, \dots, C_k\}$ (dataset of instances belonging to k classes)
- Distribution probability $(p_1, p_2, \dots, p_k)$, $p_i = \text{card}(C_i) / \text{card}(D)$
- Let A be an attribute and $v_1, v_2, \dots, v_{m_A}$ the set of values taken by this attribute
- Let $D_j = \text{set of instances from } D \text{ for which attribute } A \text{ has the value } v_j$ and P_j the distribution of D_j in the k classes ($C_{ji} = \text{number of instances having the value } v_j \text{ for attribute } A \text{ which belong to class } C_i$)
- **Information Gain** obtained by splitting the dataset according to attribute A

$$IG(D, A) = H(D) - \sum_{j=1}^{m_A} P(D_j | A = v_j) H(D_j | A = v_j), \quad H(D) = - \sum_{i=1}^k p_i \log p_i$$

$$H(D_j | A = v_j) = - \sum_{i=1}^k p_{ij} \log p_{ij}, \quad p_{ij} = \frac{\text{card}(C_{ji})}{\text{card}(D_j)}$$

$$P(D_j | A = v_j) = \frac{\text{card}(D_j)}{\text{card}(D)}$$

Choosing the splitting attribute

Example

Relation: weather.symbolic					
No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

- Class distribution (C_1 ="yes", C_2 ="no"):
- $p_1=9/14$, $p_2=5/14$
- $H(p_1, p_2)=0.94$

Outlook

	C1 (yes)	C2(no)	Frequency
Sunny	2/5	3/5	5/14
Overcast	4/4	0/4	4/14
Rainy	3/5	2/5	5/14

$$H(\text{sunny}) = -2/5 \cdot \log(2/5) - 3/5 \cdot \log(3/5) = 0.97$$

$$H(\text{overcast}) = -1 \cdot \log(1) - 0 = 0$$

$$H(\text{rainy}) = -3/5 \cdot \log(3/5) - 2/5 \cdot \log(2/5) = 0.97$$

$$IG(\text{outlook}) = 0.94 - 5/14 \cdot 0.97 - 4/14 \cdot 0 - 5/14 \cdot 0.97 = 0.94 - 0.69 = 0.25$$

Choosing the splitting attribute

Example

Relation: weather.symbolic					
No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

- Class distribution (C_1 ="yes", C_2 ="no"):
- $p_1=9/14$, $p_2=5/14$
- $H(p_1, p_2)=0.94$

Temperature

	C1 (yes)	C2(no)	Frequency
Hot	2/4	2/4	4/14
Mild	4/6	2/6	6/14
Cool	3/4	1/4	4/14

$$H(\text{hot}) = -2/4 \cdot \log(2/4) - 2/4 \cdot \log(2/4)$$

$$H(\text{mild}) = -4/6 \cdot \log(4/6) - 2/6 \cdot \log(2/6)$$

$$H(\text{cool}) = -3/4 \cdot \log(3/4) - 1/4 \cdot \log(1/4)$$

$$IG(\text{temperature}) = 0.03$$

Choosing the splitting attribute

Example

Relation: weather.symbolic					
No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

- Class distribution (C_1 ="yes", C_2 ="no"):
- $p_1=9/14$, $p_2=5/14$
- $H(p_1, p_2)=0.94$

Humidity

	C1 (yes)	C2(no)	Frequency
High	3/7	4/7	7/14
Normal	6/7	1/7	7/14

$$H(\text{high}) = -3/7 \cdot \log(3/7) - 4/7 \cdot \log(4/7)$$

$$H(\text{normal}) = -6/7 \cdot \log(6/7) - 1/7 \cdot \log(1/7)$$

$$IG(\text{humidity}) = 0.15$$

Choosing the splitting attribute

Example

Relation: weather.symbolic					
No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

- Class distribution (C_1 ="yes", C_2 ="no"):
- $p_1=9/14$, $p_2=5/14$
- $H(p_1, p_2)=0.94$

Windy

	C1 (yes)	C2(no)	Frequency
False	6/8	2/8	8/14
True	3/6	3/6	6/14

$$H(\text{false}) = -6/8 \cdot \log(6/8) - 2/8 \cdot \log(2/8)$$

$$H(\text{true}) = -3/6 \cdot \log(3/6) - 3/6 \cdot \log(3/6)$$

$$IG(\text{windy}) = 0.05$$

Choosing the splitting attribute

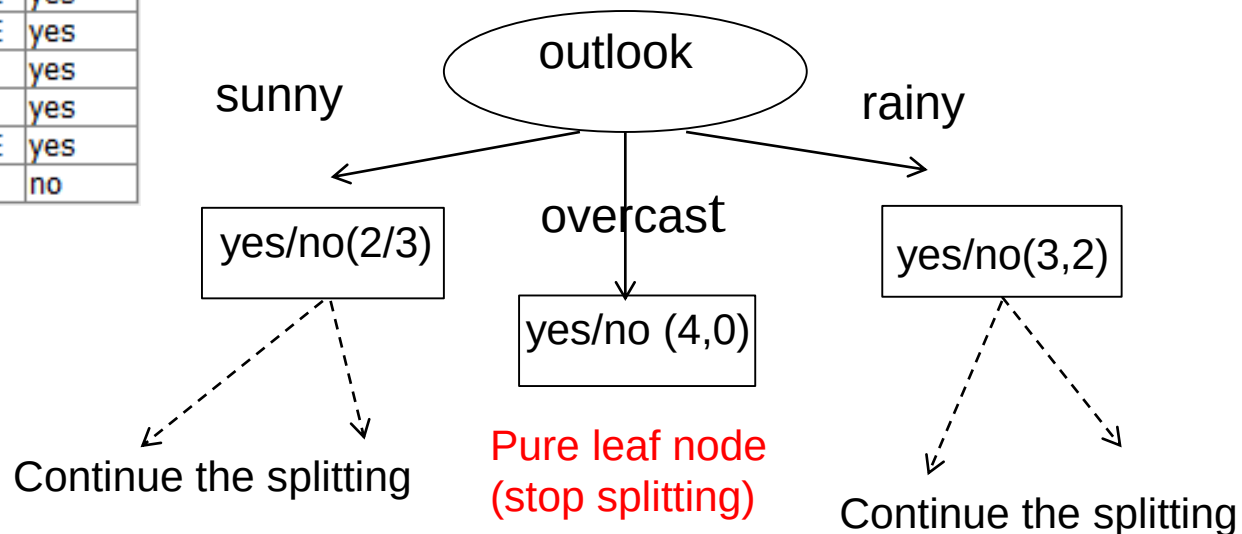
Example

Relation: weather.symbolic					
No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

Information gain for each attribute:

- $IG(\text{outlook})=0.25$
- $IG(\text{temperature})=0.03$
- $IG(\text{humidity})=0.15$
- $IG(\text{windy})=0.05$

First splitting attribute: **outlook**

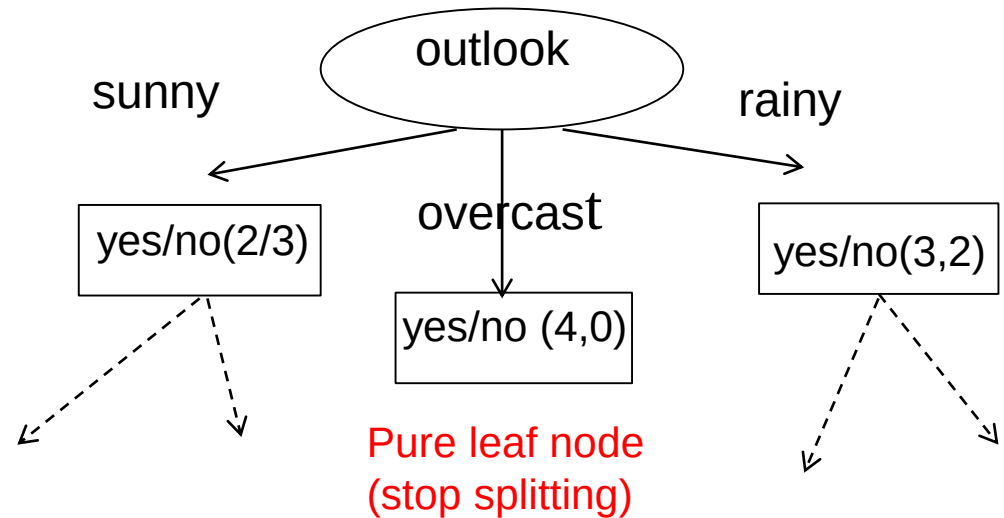


Choosing the splitting attribute

Example

Relation: weather.symbolic

No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no



Temperature

	C1 (yes)	C2(no)	Freq.
Hot	0/2	2/2	2/5
Mild	1/2	1/2	2/5
Cool	1/1	0/1	1/5

Information gain for each remaining attribute:

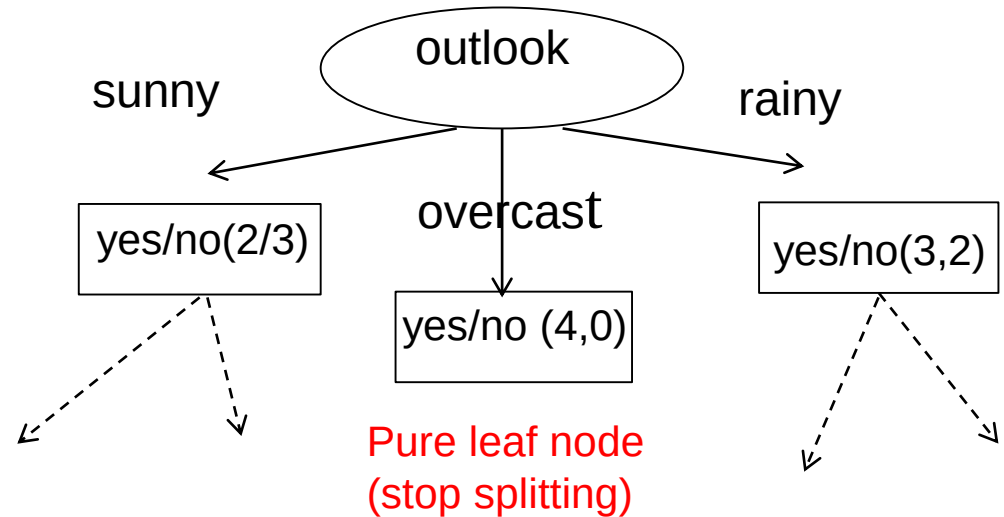
- Entropy for class distribution on “sunny” subset:
 $H(D(\text{sunny})) = -2/5 \cdot \log(2/5) - 3/5 \cdot \log(3/5) = 0.97$
- $H(\text{hot})=0$, $H(\text{mild})=1$, $H(\text{cool})=0$
- $IG(\text{temperature}) = 0.97 - 2/5 = 0.57$

Choosing the splitting attribute

Example

Relation: weather.symbolic

No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no



Humidity

	C1 (yes)	C2(no)	Freq.
High	0/3	3/3	3/5
Normal	2/2	0/2	2/5

Information gain for each remaining attribute:

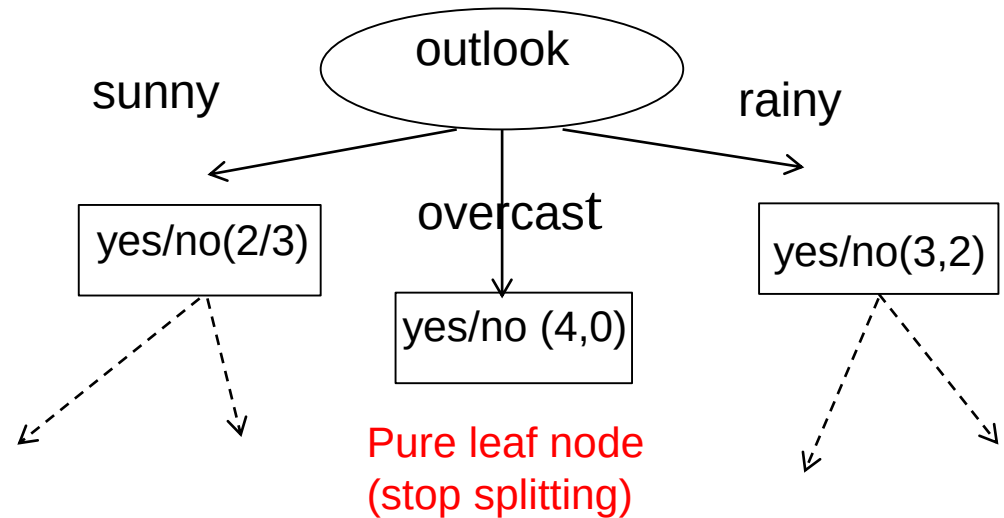
- Entropy for class distribution on “sunny” subset:
 $H(D(\text{sunny})) = -2/5 \cdot \log(2/5) - 3/5 \cdot \log(3/5) = 0.97$
- $H(\text{high}) = 0$, $H(\text{normal}) = 0$
- $IG(\text{humidity}) = 0.97 - 0 = 0.97$

Choosing the splitting attribute

Example

Relation: weather.symbolic

No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no



Windy

	C1 (yes)	C2(no)	Freq.
false	1/3	2/3	3/5
true	1/2	1/2	2/5

Information gain for each remaining attribute:

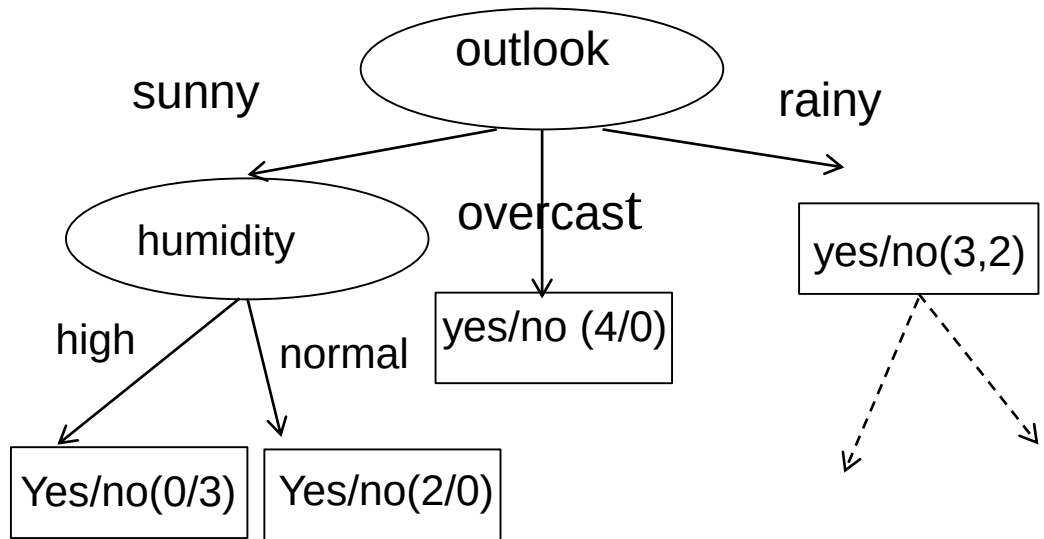
- Entropy for class distribution on “sunny” subset:
 $H(D(\text{sunny})) = -2/5 \cdot \log(2/5) - 3/5 \cdot \log(3/5) = 0.97$
- $H(\text{false}) = 0$, $H(\text{true}) = 1$
- $IG(\text{windy}) = 0.97 - 0.95 = 0.02$

Choosing the splitting attribute

Example

Relation: weather.symbolic

No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no



Pure leaf nodes
(stop splitting)

Information gain for each remaining attribute:

- $IG(\text{temperature}) = 0.97 - 2/5 = 0.57$
- $IG(\text{humidity}) = 0.97 - 0 = 0.97$
- $IG(\text{windy}) = 0.97 - 0.95 = 0.02$

Choosing the splitting attribute

Remarks:

- Information Gain favors the attributes characterized by a larger number of values
- In order to avoid this bias it can be used the **Gain Ratio**:

$$\text{GainRatio}(D, A) = \frac{IG(D, A)}{H(p_1^A, p_2^A, \dots, p_{m_A}^A)}$$

$$p_j^A = \frac{\text{card}(D, A = v_j)}{\text{card}(D)} \quad (\text{ratio of data having value } v_j \text{ for attr. } A)$$

- The splitting attribute can be selected by using the **Gini impurity measure** (smaller values are better)

Choosing the splitting attribute

Notations:

$A_1, A_2, \dots, A_n$ - attributes, $C_1, C_2, \dots, C_K$ - classes to which the data belong
 $v_{i1}, v_{i2}, \dots, v_{ir}$ - possible values of attribute i (it works only for attributes with discrete values; r_i is the number of values of attribute A_i)

Gini index for attribute A_i

$$G(A_i) = \frac{1}{N} \sum_{j=1}^{r_i} n_{ij} G(v_{ij}), \quad G(v_{ij}) = 1 - \sum_{k=1}^K p_{ijk}^2$$

n_{ij} = number of instances in the data set for which the value of A_i is v_{ij}

$$p_{ijk} = \frac{\text{number of instances in } C_k \text{ with } A_i = v_{ij}}{\text{number of instances with } A_i = v_{ij}}$$

Interpretation: the attribute with the smallest value of $G(A_i)$ is selected

Algorithms for Decision Tree Induction

ID3 (Iterative Dichotomiser):

- **Input:** dataset D
- **Output:** decision tree (input nodes labelled with attributes, leaf nodes labelled with classes, edges labelled with attribute values)

```
DTinduction (DT, D, N) /* D=dataset, DT=decision tree, N=node */
  find the best splitting attribute  $A$ 
  label node  $N$  with  $A$ 
  construct the splitting predicates (branches) for  $N$ 
  FOR each branch  $i$  of  $N$  DO
    construct the corresponding data set  $D_i$ 
    create a new child node  $N_i$ 
    IF <stopping condition>
      THEN label  $N_i$  with the dominant class in  $D_i$  ( $N_i$  is a leaf node)
    ELSE  Dtinduction (DT,  $D_i$ ,  $N_i$ )
```

Algorithms for Decision Tree Induction

C4.5 = improvement of ID3 with respect to

- **Implicit treatment of continuous attributes:**
 - incorporates a discretization procedure for continuous attributes
- **Implicit treatment of missing values:**
 - During the induction process the instances with missing values are ignored
 - During the classification the missing values of the instance to be classified are imputed
- **Splitting attribute:**
 - It uses the Gain Ratio as attribute selection (in order to be more robust with respect to the number of values)
- **Pruning:**
 - Some subtrees are replaced with leaf nodes (if the classification error is not increased significantly) – bottom-up approach

Remark:

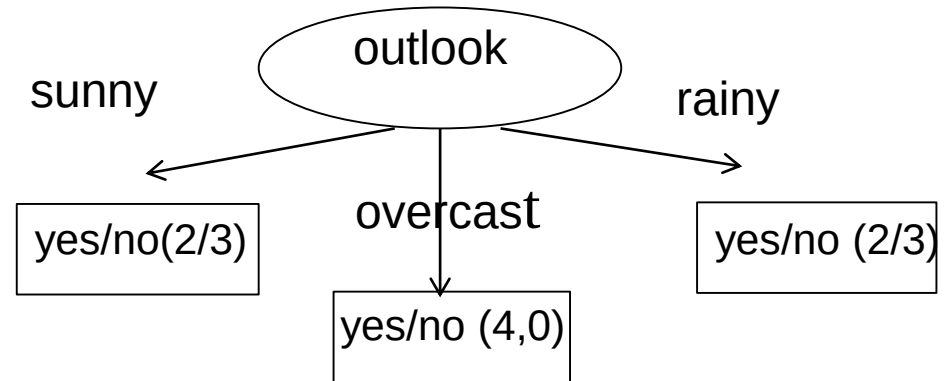
C5.0 – commercial variant of C4.5

J48 – Weka implementation of C4.5 (open source)

Algorithms for Decision Tree Induction

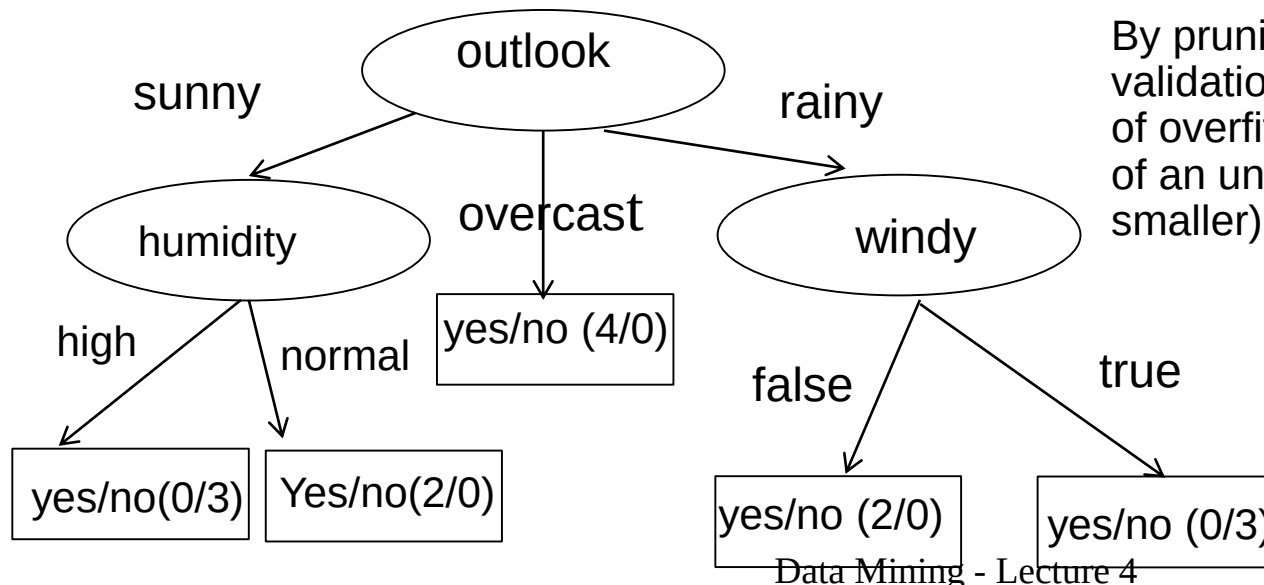
Pruning:

- Some subtrees are replaced with leaf nodes (if the classification error is not increased significantly) – bottom-up approach



Unpruned tree: error = 0

Pruned tree: error = 4/14



By pruning, the error on the training/validation set is increased but the risk of overfitting could be decreased (error of an unseen testing set could be smaller)

Classification rules induction

Reminder: classification rules are **IF ... THEN** statements containing:

- In the **antecedent** part (left hand side): conditions on the attribute values (it could be a condition concerning only one attribute or logical expressions involving several attributes)
- In the **consequent** part (right hand side): a class label

Example:

IF outlook=sunny THEN play=no

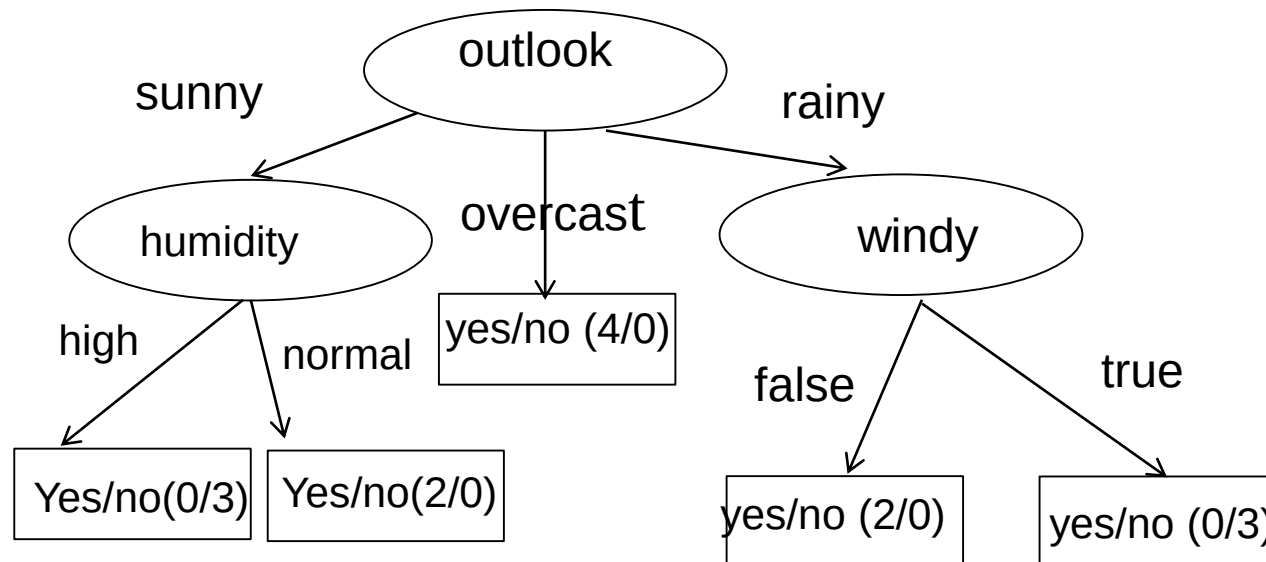
IF outlook=rainy THEN play=no

IF outlook=overcast THEN play=yes

Classification rules induction

- If the rules are extracted from a decision tree – each branch of the tree leads to a rule
- The conditions related to nodes on the same branch should be combined by AND : IF (outlook=sunny) and (humidity=high) THEN play=no
- Rules corresponding to different branches but leading to the same consequent part (same class label) can be merged by using OR between the antecedent parts:

IF (outlook=sunny and humidity=high) OR (outlook=rainy and windy =true) THEN play=no



Classification rules induction

The classification rules can be extracted directly from the data during a learning process by using covering algorithms

Notions:

- A rule **covers** a data instance if the values of the attributes match the antecedent part of the rule
- Similarly, a data instance **triggers** a rule if the values of the attributes match the antecedent part of the rule
- **Ruleset** = set of rules
- **Support of a rule** = fraction of the dataset which is covered by the rule and belong to the same class = $|\text{cover}(R) \cap \text{class}(R)| / |D|$
- **Confidence of a rule** = fraction of data instances covered by a rule which have the same class as the rule = $|\text{cover}(R) \cap \text{class}(R)| / |\text{cover}(R)|$

$\text{cover}(R)$ = set of the instances covered by R (matching the antecedent)

$\text{class}(R)$ = set of instances having the same class as R

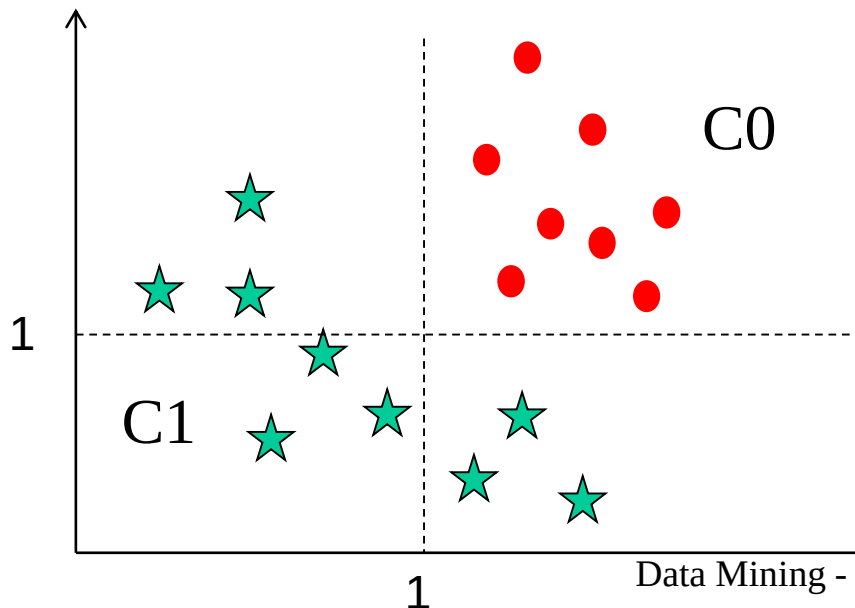
D = dataset

Classification rules induction

Notions:

- **Mutually exclusive rules** = the regions covered by rules are disjoint (an instance triggers only one rule)
- **Exhaustive ruleset** = each instance triggers at least one rule

Remark: if a ruleset is both exhaustive and the rules are mutually exclusive then it is easy to take a decision for a given instance



Example:

R1: IF $x > 1$ and $y > 1$ THEN C0

R2: IF $x \leq 1$ THEN C1

R3: IF $x > 1$ and $y \leq 1$ THEN C1

What about the case when these properties are not satisfied?

Classification rules induction

Remark: if the rules are not mutually exclusive then may appear **conflicts** (one data instance triggers several rules which have different classes)

The **conflicts** can be solved in one of the following ways:

- The **rules are ordered** (based on a specific criterion) and the decision is taken according to the first rule triggered by the data instance (rule which matches to the instance)
- The **sorting criteria** can be related to:
 - the rule **quality** (e.g. high confidence) – higher confidence is better
 - the rule **specificity** – the rules are considered better if they are more specific (e.g. those corresponding to rare classes)
 - the rule **complexity** (e.g. number of conditions in the antecedent part) – simpler rules are considered better

Remark: these criteria can be conflicting (a high confidence rule is not necessarily a simple rule)

- The result is the **dominant class** from the set of rules triggered by the data instance

Classification rules induction

Sequential covering algorithm:

Input: data set

Output: ordered set of rules

Step 1: Select a class label and determine the “best” rule which cover the data instances from D having the selected class label. Add this rule to the list of rules

Step 2: Remove all data from D which match the antecedent of the added rule. If there are still class labels to select and data in D go to Step 1

Remark:

- This is the general structure of sequential covering algorithms
- Particular algorithms differ with respect to the class selection strategy
- Unlike the decision trees induction algorithms that rely on a divide-and-conquer strategy, the covering algorithms rely on a separate-and-conquer strategy

Classification rules induction

Example: RIPPER (Repeated Incremental Pruning to Produce Error Reduction)
[W. Cohen, “Fast Effective Rule Induction.”, *Machine Learning Proceedings 1995*]

Particularities:

- **Class-based ordering**: the classes are selected in order of their size (the small classes are selected first)
- The rules corresponding to one class are placed contiguously in the ordered list of rules
- The addition of a new rule corresponding to one class is stopped:
 - when the rule becomes too complex
 - when the new rule has a classification error (on a validation set) which is larger than a predefined threshold
- If at the end some uncovered data remain then is defined a “**catch all**” rule to which is assigned the **dominant class**

Classification rules induction

Example: RIPPER (Repeated Incremental Pruning to Produce Error Reduction)

Details: incremental construction of a rule (it is supposed that the target class is given and fixed)

- Starting rule: $\{\} \Rightarrow C$
- The antecedent is increased by adding one conjunct (i.e. term in the antecedent) at a time (e.g. the rule becomes $(A_i=v_{ij}) \Rightarrow C$) using the FOIL (First Order Induction Learner) information gain of transforming rule R_0 in rule R_1 (by adding a new term)

$$IG(R_0, R_1) = p_1 * (\log(p_1/t_1) - \log(p_0/t_0))$$

- p = relative frequency of instances covered by the rule which are from the target class C (p_1 corresponds to R_1 and p_0 corresponds to R_0)
- t = relative frequency of instances covered by the rule, disregarding the class (t_1 corresponds to R_1 and t_0 corresponds to R_0)
- The term leading to the largest gain is preferred

Summary

- Construction of classification models
 - It requires labelled data (supervised training)
 - Dataset split: training subset + validation subset + testing subset
- Performance measures
 - Confusion matrix (true positives/negatives, false positive/negatives)
 - Accuracy, sensitivity (recall), specificity, precision
 - Aggregation of measures: F-score, AUC (area under the receiver operating curve)
- Algorithms:
 - Decision trees: recursive splitting (ID3, C4.5)
 - Classification rules: covering algorithms - incremental construction (RIPPER)
- Interpretable models: decision trees, classification rules
 - <https://christophm.github.io/interpretable-ml-book/tree.html>
 - interpretable models: <https://github.com/csinva/imodels>
 - dedicate library: <https://github.com/csinva/imodels>
 - Details on Rule Learning: <https://ibook.pub/ql/foundations-of-rule-learning>

Next lecture

- Instance based classifiers
 - kNN – nearest neighbor
- Probabilistic Classification
 - Naïve Bayes